

WELL COMPLETION REPORT (SYNTHETIC)

OIL-HPJ-003 - Hapjan-003

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000031
Field / District	Hapjan / Tinsukia
Block	OALP-III-07
Basin	Assam-Arakan
Formation	Girujan (GRJ)
Spud / Completion	2004-10-30 / 2005-02-25
Measured / TVD	1968 m / 1941 m
Trajectory	Directional; inc 26.9 deg; az 115.1 deg
Formation Interval	955-1518 m
Lithology	Claystone-sandstone alternation
Pore Pressure Reference	1949 psi
Fracture Gradient	14.16 ppg
Mud System	Low-Solids WBM
NWIS Well Risk	Moderate (52.0/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	78	Conductor
13 3/8 in casing	492	Surface
9 5/8 in casing	1141	Intermediate
7 in liner/casing	1731	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2004-11-14 360 m	Stuck Pipe	Critical	Hookload became erratic during pipe movement. Mitigation: Reduce ROP and increase circulation before high-angle interval. Lesson: Avoid prolonged static periods in high-drag interval.
2004-12-19 935 m	Cementing Issue	Critical	Displacement pressure became unstable. Mitigation: Verify cement placement using pressure and volume balance. Lesson: Plan conservative displacement rate in weak formations.
2004-11-19 453 m	Kick	High	Flow-out exceeded flow-in and gas units increased. Mitigation: Shut in, verify influx and circulate using approved well-control procedure. Lesson: Escalate connection-gas trends before entering correlated pressure zone.
2004-11-21 486 m	Mud Loss	High	Partial returns observed at shale shaker. Mitigation: Reduce ECD, treat losses with LCM and confirm stable returns. Lesson: Use conservative hydraulics when re-entering equivalent loss-prone interval.
2004-11-27 577 m	Torque Spike	High	Sharp torque excursion occurred with RPM instability. Mitigation: Use staged parameter reduction to avoid torsional oscillation. Lesson: Avoid simultaneous high WOB and high RPM in correlated interval.
2005-01-28 1557 m	Stuck Pipe	High	Rotary movement deteriorated progressively. Mitigation: Reduce ROP and increase circulation before high-angle interval. Lesson: Use offset stuck-pipe depth to increase circulation and reaming frequency.
2005-02-06 1683 m	NPT	High	ROP dropped to zero during equipment unavailability. Mitigation: Maintain critical spares for recurring equipment failures. Lesson: Separate geological risk from equipment-driven NPT in analytics.

Completion Summary

The synthetic well reached 1968 m MD in the Girujan interval. The NWIS knowledge record links 12 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.